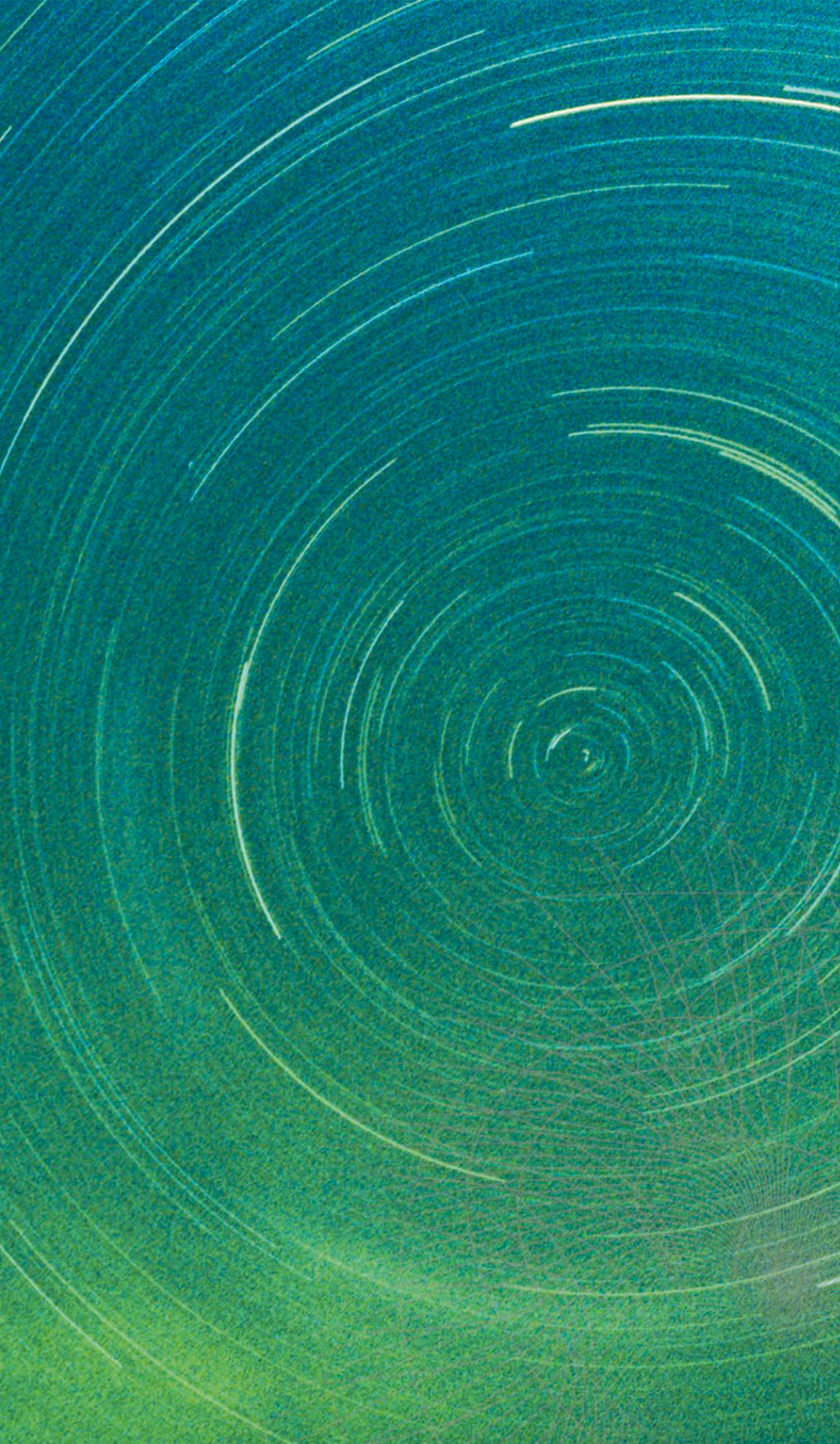




*“Mortal as I am, I know that I am born for a day.  
But when I follow at my pleasure the serried  
multitude of the stars in their circular course, my  
feet no longer touch the Earth.”*

Ptolemy

OBJECTS IN THE UNIVERSE—galaxies, stars, planets, nebulae—are scattered across three dimensions of space and one of time. Viewed from widely separated locations in the universe, their relative positions look completely different. To find objects in space, study their movements, and make celestial maps, astronomers need an agreed reference frame, and for most purposes the frame used is Earth itself. The prime element of this Earth-based view is the celestial sphere—an imaginary shell around Earth to which astronomers pretend the stars are attached. Apparent movements of celestial objects on this sphere can be related to the actual movements of Earth, the planets (as they orbit the Sun), the Moon (as it orbits Earth), and the stars as they move within the Milky Way. Understanding the celestial sphere, and conventions for naming and finding objects on it, are essential first steps in astronomy.

**MOVEMENT ON THE SKY**

This photograph, obtained over a four-hour period from the Las Campanas Observatory in Chile, looks toward the south celestial pole. The circular, clockwise star trails across the sky are a feature of the Earth-based view of the cosmos, since they result solely from the Earth's rotation.

# THE VIEW FROM EARTH

